MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, ELECTROLYTIC (NONSOLID ELECTROLYTE), TANTALUM, (POLARIZED, HIGH ETCHED FOIL), 85°C (VOLTAGE DERATED TO 125°C), ESTABLISHED RELIABILITY, STYLE CLR71

This amendment forms a part of Military Specification Sheet MIL-C-39006/16A, dated 30 January 1976, and is approved for use by all Departments and Agencies of the Department of Defense.

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* Following preamble, add:

CAUTIONARY NOTE: THESE CAPACITORS MAY EXPLODE IF SUBJECTED TO REVERSE VOLTAGES. WHEN FIRST TESTED FOR DC LEAKAGE OR AT OTHER HIGH POTENTIALS, EXERCISE CAUTION TO VERIFY CORRECT POLARITY MARKINGS.

FAILURE RATE LEVEL 'L' INACTIVE FOR DESIGN AFTER 27 May 1976

CASE SIZES E3, E4, and E5 INACTIVE FOR DESIGN AFTER 27 May 1976

CASE SIZE E1 INACTIVE FOR DESIGN AFTER 20 Oct 1978

CASE SIZE E2 INACTIVE FOR DESIGN AFTER 12 May 1980

FIGURE 1 (dimension table), C Max, Case size G2,: Delete "1.406 (35.71)" and substitute "1.469 (37.31)".

FIGURE 1 (dimension table): Add G1 case size

G1	.688 (17.48)	.188 (4.78)	.219 (5.56)	1.188 (30.18)

Constant voltage conditioning: Delete "MIL-C-3906/" and substitute "MIL-C-39006."

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TABLE I, following part no. M39006/16-: Add "1/".

TABLE I, part number column: Add "-" following numbers 2036 through 2174 to denote optional random vibration part numbers.

TABLE I: Delete line 14 in its entirety. For replacement use part numbers M39006/01-3200, -3201-, -3203-, -3204-, and -3205-.

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TABLE I: Add the new G1 case size units.

Rated voltage (85°C)	Cap.	Cap. tolerance	DC leakage (max) 25°C	DC leakage (max) 85°C	DC leakage (max) 125°C	Derated voltage (125°C)	Surge voltage (85°C)	Impedance (max)	Case size	Part no. M39006/16- 1/ Failure rate level (%/1,000 hr) L (2.0)	M (1.0)	P (0.1)	R (0.01)	S (0.001)
15	29	+75,-15	7	35	23	10	17.2	137	G1	2000	2035-	2070-	2105-	2140-
25	17	+75,-15	6	34	21	15	28.8	220	G1	2005	2040-	2075-	2110-	2145-
30	15	+75,-15	6	34	22	20	34.5	250	G1	2010	2045-	2080-	2115-	2150-
50	7	+50,-15	5	28	17	30	57.5	550	G1	2015	2050-	2085-	2120-	2155-
75	4	+50,-15	5	24	16	50	86.2	930	G1	2020	2055-	2090-	2125-	2160-
100	3	+30,-15	4	20	14	65	115.0	1210	G1	2025	2060-	2095-	2130-	2165-
150	1.6	+30,-15	3	18	12	100	172.0	2340	G1	2030	2065-	2100-	2135-	2170-

Following Table I: Add "1/ Dash number for failure rate level M, P, R, and S parts will include the letter 'H' to indicate the optional random vibration requirement of '-' will be deleted."

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Following vibration requirement, add new Random vibration requirement:

"Random vibration (when applicable): Method 214 of MIL-STD-202, test condition II-E (18.7 g)."

Temperature cycling: Delete "Temperature cycling: Method 102 of MIL-STD-202, condition D" and substitute "Thermal shock: Method 107 of MIL-STD-202, condition A."

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Life at 85°C: Delete "240-hr (group B)" requirement in its entirety; also delete "10,000-hr (group C)" and substitute "10,000-hr (group B)".

Vacuum life: Delete "DF" requirement and substitute "DF" - 20% (≤50 V) or 15% (>50 V) (max)."

Add the following new requirement under "Life at 125°C":

"10,000-hr (group B):

DCL (125°C) - 125% (max) of table I value.

DCL (25°C) - See table I.

ΔCap. - Within ±50% of initial measurement for E1, E2, and G1 cases; within ±25% of initial measurement for all other cases.

DF - 80% (≤30 V) or 60% (>30 V) (max) for E1, E2, and G1 cases; 40% (≤30 V) or 30% (>30 V) (max) for all other cases.

DWV - 2,000 V dc, min.

IR - 100 megohms, min."

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NOTE: The margin of this amendment is marked with an asterisk to indicate where a change (additions, modifications, corrections, deletions) from the previous amendment was made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
Army - ER
Navy - EC
Air Force - 11

Review activities:
Army - AR
Air Force - 17, 85, 99
DLA - ES

User activities:
Navy - AS, MC, OS
Air Force - 19

Agent:
DLA - ES

Preparing activity:
Navy - EC

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